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Linear stability analysis of a liquid jet in a weak crossflow

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In an effort to better understand the primary breakup mechanism of liquid jets in crossflow, a theoretical model for three-dimensional spatial linear stability of a viscous liquid jet injected in a weak gaseous crossflow is developed. The results are presented for the behavior of jet instabilities in the range of crossflow-to-jet momentum flux ratio of less than 0.01. In this range, the deformation of the liquid column allows a parallel base flow which is described by a modified viscous potential flow analysis. Focusing on the early stages of the jet evolution, a characteristic equation accounting for the growth of columnar waves is obtained and the most dominant wavelength and the corresponding growth rates are calculated. The predicted results for the asymptotic case of free jet flow are examined against the results available in the literature. Symmetric and asymmetric modes of liquid jet disturbance are investigated for a wide range of viscous, surface tension, and aerodynamic force ratios. *Published by AIP Publishing.* https://doi.org/10.1063/1.5043589

I. INTRODUCTION

A liquid jet injected transversely into a gas flow enhances the momentum exchange between two streams. Taking this advantage, the breakup of liquid jets in crossflow is a critical process in numerous engineering systems including fuel injection and thermal spraying.[1] Insight concerning the mechanisms of primary breakup of liquid jets in crossflow can be gained by studying the properties of waves that appear along the surface of the jet during the breakup process. In fact, a stream of a gas flowing over a stream of a liquid causes instabilities at the liquid-gas interface. The growth of these instabilities leads to disintegration of the liquid volume into smaller droplets, which enhances the mixing of the liquid and the gas.

After injection, the liquid column is deflected in the direction of the crossflow and its cross section undergoes deformation (Fig. 1). This behavior is caused by acceleration of the gas across the liquid jet and pressure variation at the jet circumference. A high pressure zone is formed at the windward surface and the gas pressure is reduced along the sides of the jet with the lateral motion of the liquid jet eventually stabilized by surface tension. The increased drag forces due to the flattened shape of the liquid jet enhance its tendency to be deflected in the crossflow direction.[2,3] The waves begin to appear along the length of the liquid jet with the wavelength equal to several liquid jet diameters and progressively reduce with increasing crossflow-to-jet momentum ratio. These waves gradually grow on the windward of the jet until sufficiently amplified to cause the jet column to breakup into ligaments. In higher gas-momentum values, shear breakup occurs consistent with the growth of azimuthal instabilities leading to the formation of undulations around the periphery of the jet that eventually disintegrate into ligaments and drops.

Basically, two types of disturbances exist on the jet surface; the axial disturbances leading to the column breakup and the azimuthal disturbances leading to the surface breakup. The formation of these waves was speculated to be, on top of the capillary instability which exists always, either as a result of Kelvin–Helmholtz (hereafter KH) instability or Rayleigh–Taylor (hereafter RT) instability.[4,5] The relative velocity between the horizontal gaseous crossflow and the vertical liquid jet, as well as the discontinuity in their properties, drives the waves on the front face and results in KH instability at the surface of the jet. Additionally, on the upstream side of the liquid jet, in the vicinity of its upstream stagnation point, crossflow tends to accelerate the gas toward the liquid jet in the cross stream direction, developing a RT instability.[4] These RT waves will finger into the liquid and corrugate the front of the column toward the jet lateral by shear flow of gas coming from the high-pressure stagnation point, causing it to break up. Liquid jets in crossflow exhibit different levels of RT and KH instabilities depending on operating conditions and inlet turbulence levels.[5–7] However, no theoretical investigation has been reported to support the interaction mechanism of KH and RT disturbances.

Although most of the research on liquid jet in a gaseous crossflow have been done experimentally by shadowgraph and high-speed imaging, for example, by Wu *et al.*,[8] Mazallon *et al.*,[9] and Sallam *et al.*,[2] experimental techniques often have difficulties visualizing the breakup mechanism at the gas–liquid interface, especially on the downstream side of the jet due to the presence of many droplets. It is also challenging to perform parametric experimental studies that isolate the effect of non-dimensional groups of viscous, surface tension, and aerodynamic force ratios related to liquid jet and crossflow. However, simple theoretical models can be developed to help explore the parameter ranges which are hard to work at in a laboratory setting.[5]

On the other hand, numerical modeling approaches have encountered severe difficulties in capturing the impact of

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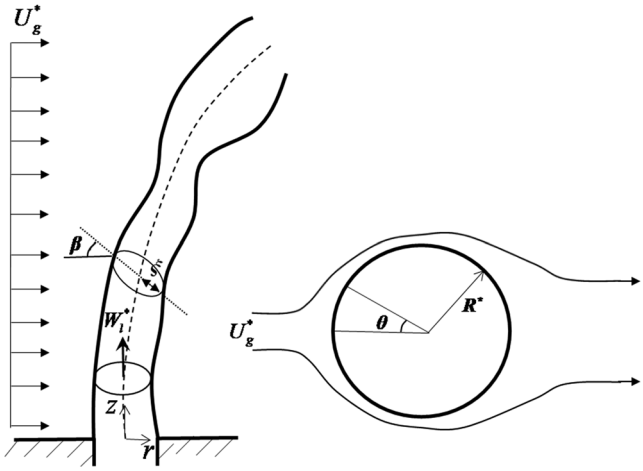

FIG. 1. Schematic of the liquid jet in crossflow and relevant quantities.

operating conditions on the detailed atomization physics. The complex multiphase and multiscale physics involved in the interaction of deformed jet and gas flow, makes the analysis of flow structures complicated. One not only has to capture the sharp moving and deforming interface but also has to account for the singular surface tension force creating the phase interface a discontinuity.[10] It is noted that simulating jet atomization at a realistic high density ratio at ambient conditions remains a grand challenge for most of the numerical methods.[7] An increase in the density ratio reduces the wavelength of disturbances. Thus a sufficient fine mesh is required to capture the growth of such small scale waves on the jet surface. Therefore, the detailed numerical simulation of this complex problem in real conditions is still computationally expensive and, therefore, low cost theoretical models need to be developed.

Theoretical studies of the breakup mechanisms along with experimental imaging techniques and computational fluid dynamics simulations can all contribute to providing estimates for the missing information at the near field. Theoretical studies can be used to obtain basic information on the deformation of the jet and the path it takes in the gas stream before breakup. Obviously, the more extreme the flow conditions become, the less capable theoretical models of these types become in capturing the details of the flow structures, and one has to rely on numerical simulations and experimental results.[6] However, using theoretical models, one can ideally explore the entire parametric space beforehand and help deciding the range of parameters over which numerical or experimental studies are to be performed.[1]

Developing the linear hydrodynamic stability analysis associated with breakup of liquid jets in coflow is well established and extensively studied, e.g., by Sterling and Sleicher,[11] Yang,[12] and Avital.[13] Although, due to turbulence and nonlinearity effects, at higher jet speeds, the value of linear stability analysis is not confirmed; it provides a qualitative description of the physical mechanisms involved in the various regimes of the jet breakup process. Actually, many of today's atomization models draw heavily from the linear theories based on a column of liquid exposed to a high-velocity gas shear flow.[1]

Although several studies have been done on linear stability analysis of single phase jets in crossflow, e.g., by Rottman *et al.*,[14] Coelho and Hunt,[15] Higuera and Martinez,[16] Alves *et al.*,[17] and Karagozian,[18] only a couple of studies have been reported on liquid jet in crossflow. A pioneer work in this area is that of Kitamura and Takahashi[19] who studied the instability of the stagnant liquid column in the transverse low inertia airflow. Their theoretical analysis is based on the assumption that disturbances in the liquid are symmetric, while those in the surrounding air are asymmetric. More recently, using a temporal azimuthal stability analysis, Behzad *et al.*[20] studied the surface break up of a non-turbulent liquid jet in a crossflow close to the nozzle exit where the jet cross section remains circular.

In spite of the fact that much work has been done experimentally and numerically on the characterization of spray, the understanding of the breakup process is far from complete and more studies of the near-field jet instabilities are needed.[21] In the near field, the instabilities initiated, grow along the jet and lead to deformation and breakup of the liquid column, in the far-field. The perturbations are initiated and grow linearly, before developing nonlinearly downstream. Therefore, linear stability analysis can be very useful at the early stages of jet evolution. In fact, linear stability analysis provides a qualitative description of the physical mechanisms involved in the various regimes of the jet breakup process. The physical mechanism of breakup frequently remains the same during nonlinear evolution, and the nonlinear evolution subsequent to the onset of breakup alters these physical mechanisms quantitatively only.

This work is aimed at exploring such instabilities toward resolving the primary breakup mechanism. To this end, using a 3D irrotational viscous flow in the near field of a weak crossflow, the mean flow is obtained first. Then by employing a normal mode decomposition of governing equations for each phase, and applying matching conditions at the interface, a linear stability analysis is carried out. Next, using the obtained dispersion relation, the spatial evolution of waves is explained and the instability dynamics is parametrically investigated. The paper finishes with conclusions.

II. PROBLEM DESCRIPTION AND GOVERNING EQUATIONS

The problem configuration that is considered in the present work and relevant parameters are schematically illustrated in Fig. 1. A non-turbulent liquid jet with density ρ_l^*, dynamic viscosity μ_l^* and surface tension σ^* is streaming with a uniform axial velocity W_l^* from a round nozzle with radius R^* through an immiscible and incompressible gas with density ρ_g^* and dynamic viscosity μ_g^* which has a transverse velocity of U_g^*. The effect of gravity is negligible in the present range. The cylindrical coordinate system, centered at the nozzle exit, is denoted by (r, θ, z) corresponding to the velocity field (U, V, W), respectively, i.e., the z-direction is aligned with the direction of liquid jet injection and the jet cross section is described by r and θ coordinates. All quantities in governing equations are non-dimensionalized by using R^*, ρ_l^*, and W_l^* where an asterisk refers to a dimensional quantity.

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The following dimensionless groups can be identified:

$$We = \frac{\rho_l^* W_l^{*2} R^*}{\sigma^*}, \quad Re = \frac{\rho_l^* W_l^* R^*}{\mu_l^*}, \quad q = \frac{\rho_g^* U_g^{*2}}{\rho_l^* W_l^{*2}}, \quad h = \frac{\mu_g^*}{\mu_l^*}, \quad l = \frac{\rho_g^*}{\rho_l^*}, \tag{1}$$

known as the Weber number, Reynolds number, (gas-to-liquid) momentum flux ratio, viscosity ratio, and density ratio, respectively. In some cases, it will be more convenient to use the Ohnesorge number defined as

$$Oh = \frac{\mu_l^*}{\sqrt{\rho_l^* \sigma^* R^*}} = \frac{We^{0.5}}{Re}. \tag{2}$$

In order to carry out an instability analysis, a base flow for the crossflow jet needs to be defined. However, no steady exact solution to the Navier-Stokes equations exist for this three-dimensional flow field. Therefore, due to the complexity of the flow field around and inside the liquid jet, in order to have progress in theoretical analysis and make it tractable, some simplifying assumptions are needed as described below.

For the near field of a strong liquid jet in a weak crossflow, the flow field can be approximated by an irrotational flow solution as developed by Joseph *et al.*[22] We note that solutions of the Navier-Stokes equation can be decomposed into a rotational part and an irrotational part. Potential flows of incompressible fluids with constant properties are irrotational solutions of the Navier-Stokes equation that satisfy Laplace's equation. The theory of irrotational flows of a viscous fluid is an approximate theory that works well especially in gas-liquid flows of liquid at low Reynolds numbers. The theory of stability of free-surface problems based on viscous potential flow was developed by Joseph *et al.*[22] and could be used as a model in course of hydrodynamic stability.[23] With that, the complete set of equations includes conservation of mass and momentum which by introducing the velocity potential ϕ reduces to Laplace's equation as

$$\nabla^2 \phi_l = 0, \quad \nabla^2 \phi_g = 0, \tag{3}$$

where subscripts l and g denote liquid and gas phases, respectively.

Analogous to the flow past a solid cylinder, the liquid jet can reasonably be represented by the following approximate potential solution:

$$\phi_l = W_l z + O(q), \tag{4}$$

and similarly, its external flow field by

$$\phi_g = \sqrt{\frac{q}{l}}\left(r + \frac{1}{r}\right)\cos\theta + O\left(q^{1.5}\right). \tag{5}$$

Velocity components are obtained from gradient of potential flow functions, and pressure is expressed from the Bernoulli's equation as

$$p_l = -\left(\frac{\partial \phi_l}{\partial z} + \frac{\partial \phi_l}{\partial t}\right) \tag{6}$$

for liquid, and

$$p_g = -l\left(\sqrt{\frac{q}{l}}\frac{\partial \phi_g}{\partial r} + \frac{\partial \phi_g}{\partial t}\right) \tag{7}$$

for gas. However, this solution has a discontinuity in the tangential and axial velocity components at the gas-liquid interface from a high gas velocity in the crossflow to zero liquid velocity in the jet and hence, it is accurate only outside the boundary layers. Although the base flow is different at the realistic boundary layer where this sharp velocity gradient on either sides of the interface is smoothened, it facilitates analytical progress while maintaining the relevant physics as in the continuous profile.

In general, high-Reynolds-number flows may be approximated by an outer irrotational flow and small layers on the boundary and narrow wakes where vorticity is important. In the present case, it is assumed that the vorticity layer is thin and its influence on the bulk motion and on the irrotational effects of viscosity is minor. Following Joseph *et al.*,[22] in Eqs. (6) and (7), a pressure correction term is added due to viscous effects on boundary layers around liquid because of extra pressure generated by the unphysical shear stress at the outer edge of the boundary layer. This pressure term for flow past a cylinder[22] is

$$p = 4\left(\frac{h}{Re}\sqrt{\frac{q}{l}}\right)\cos\theta. \tag{8}$$

Employing the analogy of flow past a cylinder eliminates the complication of involving the axial variations of the base flow in the stability analysis and allows for a theoretical progress by reducing the problem to a simple parallel basic flow. In order to use this analogy, it has to be shown that for a weak crossflow, the base flow varies slowly in the axial direction compared to the perturbed flow, i.e., the base flow velocity W_l^* varies in the axial direction on a scale much larger than the wavelengths of axial disturbances, λ, which grow on the interface. To consider this scale-disparity, a length-scale ratio δ is introduced,[24,25] which compares the characteristic scales of the instability L_{inst} to the characteristic length of the mean-flow deformation L_{mean}, with the following equation:

$$\delta = \frac{L_{inst}}{L_{mean}}. \tag{9}$$

To assess the validity of the current perturbation analysis, δ is evaluated from relation (9) over the range of operating conditions that are considered in this work. Using a theoretical analysis based on time integration of the momentum equation, it has been shown[26] that trajectory of a liquid jet centerline in crossflow follows this equation,

$$r = \left(\frac{C_d q}{2\pi}\right) z^2, \tag{10}$$

where C_d is the drag coefficient. In the present analysis, L_{inst} is associated with the perturbation wavelength, on the order of one jet diameter, and L_{mean} is set as the axial distance over which the jet is horizontally deflected by one jet radius. With this definition,

$$L_{mean} = \left(\frac{\pi}{C_d}\right)\frac{1}{qz}, \tag{11}$$

and therefore,

$$\varepsilon = \left(\frac{2C_d}{\pi}\right) qz. \tag{12}$$

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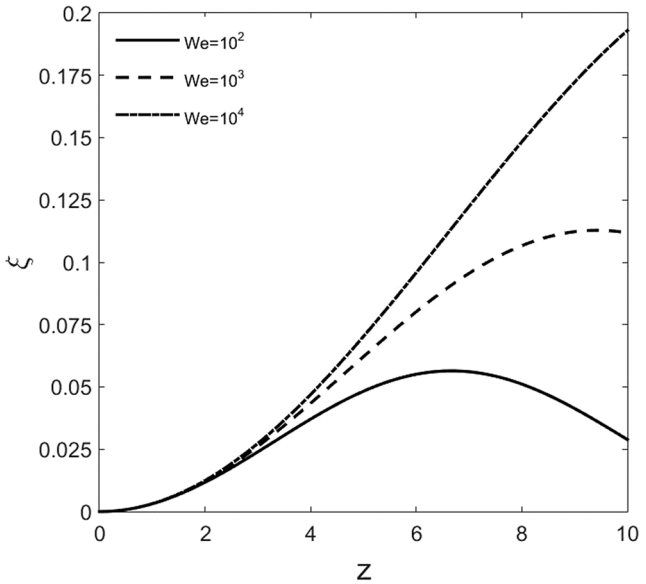

FIG. 2. Deformation of jet cross section along its axis for $q = 0.01$ and $Oh = 0.0042$.

Based on Eq. (12), and assuming $C_d = 1.67$,[26] the length-scale ratio δ remains less than 0.1 for $q \leq 0.01$, if $z \leq 10$. In this range deformation is negligible, justifying the parallel mean flow approximation, in which streamwise variations of the base flow are not taken into account in the stability analysis.

Observations of Hsiang and Faeth[27] and Wu *et al.*[8] suggested qualitative similarities between the primary breakup regimes of liquid jets in gaseous crossflow and the secondary breakup of drops subjected to shock-wave disturbances. Based on a mass-spring-damper analogy for droplet deformation,[28] the following equation for deformation of jet cross section can be proposed:[26]

$$\left(\frac{\xi^2}{4}\right)\xi_{zz} + \left(\frac{1}{Re}\right)\xi_z + \left(\frac{s}{We}\right)\xi = \left(\frac{\xi\cos^2\beta}{3\pi^2}\right)q, \qquad (13)$$

where ξ is the deviation of center of mass of the half-cross-section from its undeformed position (Fig. 1). In this formula, β is the angle of deflection of the liquid column equal to the complement of slope of trajectory in Eq. (10) as shown in Fig. 1. In Eq. (13), s is a weak function of ξ almost equal to unity. This equation is solved numerically using a fourth-order Runge-Kutta method with initial conditions of $\xi(0) = \xi_z(0) = 0$. The variation of ξ for $q = 0.01$, $Oh = 0.0042$ and various We numbers is shown in Fig. 2. We note that a water jet exiting from a nozzle with diameter of 1 mm is associated with $Oh = 0.0042$. Results confirm that in the near-field region for cases with low crossflow-to-jet momentum flux ratios, which are of primary concern in this study, the jet bending and deformation of its cross section is negligible and jet preserves its nearly cylindrical shape.

A numerical solution will give us an idea about the validity of assigning the above simplified set of equations. To this end, taking advantage of a finite volume solver, the equations of Navier-Stokes and continuity are solved numerically for air-water with constant density and viscosity and ignorable body force. The equations of conservation of mass and momentum

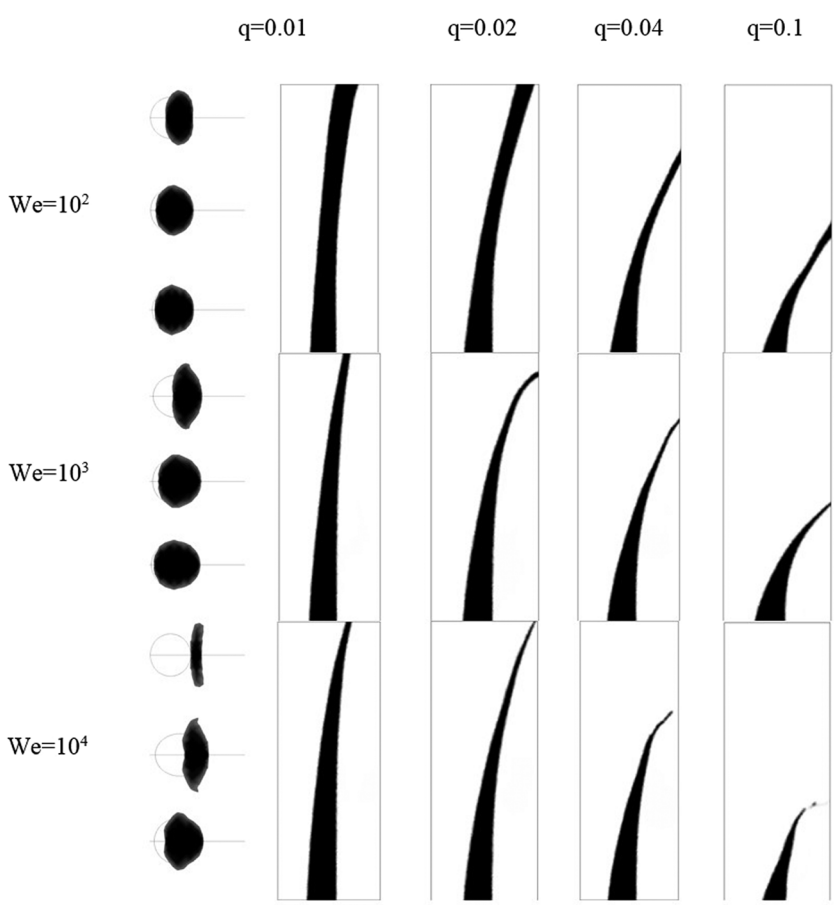

FIG. 3. Deformation of jet trajectory and its cross section for various We and q values at $Oh = 0.0042$. Jet cross sections are shown only for $q = 0.01$ at $z = 2, 4, 10$. Jet and crossflow are streaming in the same direction as Fig. 1.

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are as follows:

$$\nabla \cdot \vec{V} = 0, \tag{14a}$$

$$\frac{\partial \vec{V}}{\partial t} + \nabla \cdot \left(\vec{V}\vec{V}\right) = -\nabla p + \frac{1}{Re}\nabla^2 V + \vec{F}. \tag{14b}$$

The surface tension force, F, on the interfacial cells is calculated as multiplication of surface tension σ^*, surface curvature κ, and the gradient of volume fraction, f. We note that f is a scalar $0 \le f \le 1$, taking zero for empty cells and unity for completely liquid-filled cells and accordingly,

$$\kappa = -\nabla \cdot \left(\frac{\nabla f}{|\nabla f|}\right). \tag{14c}$$

To capture the gas-liquid interface, in addition to the above equations, a volume fraction advection equation should be solved for both liquid and gas phases as

$$\frac{\partial f}{\partial t} + \nabla \cdot \left(f\vec{V}\right) = 0. \tag{14d}$$

Equations (14a)–(14d) are solved numerically for both liquid and gas phases.

The open source multiphase flow solver InterFoam is used to solve the laminar 3D Navier-Stokes equations. Taking advantage of a finite volume method, the advection terms in the discretized conservation equations are solved using a second order scheme. The phase distribution itself is calculated using the volume of the fluid model to identify the liquid-gas interface. The grid needs to be finer near the interface to capture the interface accurately. The grid independence checking was made for a finer mesh and the results of jet deformation show that for less than the minimum cell size of 1×10^{-5} m adjacent to the interface, the difference between the results is negligible.

Figure 3 shows part of the numerical results depicting trajectory and cross section of a liquid jet in different q and *We* values. In addition, distribution of pressure at the interface is shown in Fig. 4, confirming the validity of analogy of flow past a cylinder for the present model. In agreement with experimental results of Wu *et al.*,[8] numerical results confirm that for q small enough and still at near field, the jet bending and deformation of its cross section is small and limited to surface disturbances. Therefore, the model of flow around a solid cylinder can be employed for the base flow around a liquid jet. Results also confirm that for high momentum ratios, the mean flow is considerably deforming and a multi-scale analysis[24,25] is required to perform the stability analysis. However, this extended analysis is beyond the scope of the current work.

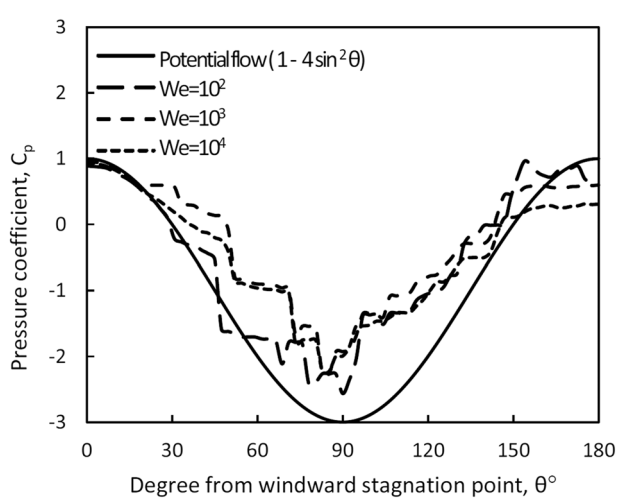

FIG. 4. Distribution of pressure at jet interface at $z = 2$, for $q = 0.01$ and $Oh = 0.0042$.

III. SPATIAL LINEAR STABILITY ANALYSIS

The analysis of the columnar instability can be performed on a 3D jet configuration sufficiently close to the nozzle exit to satisfy the above mentioned assumptions. In this section, a perturbation approach is used to study the governing equations for wave propagation on the jet surface. For a linear stability analysis, an instantaneous flow-field quantity ψ is decomposed into a base flow component $\overline{\psi}$ ψ and infinitesimal disturbances denoted by ψ' as

$$\psi(r,\theta,z,t) = \bar{\psi}(r,\theta,z) + \psi'(r,\theta,z,t). \tag{15}$$

Following the classic normal mode analysis, the disturbances are expanded into Fourier modes through

$$\psi'(r,\theta,z,t) = \sum_{m=-\infty}^{\infty} \widehat{\psi}_m(r) e^{im\theta} e^{i(kz-\omega t)}, \tag{16}$$

with

$$\widehat{\psi}_m(r) = \frac{e^{-i(kz-\omega t)}}{2\pi} \int_0^{2\pi} \psi' e^{-im\theta} d\theta, \tag{17}$$

where m is the azimuthal wavenumber and ω is the Strouhal number, related to the real-valued frequency ω^* by

$$\omega = \frac{\omega^* R^*}{W_l^*}. \tag{18}$$

In addition, $k = k^* R^*$ is complex-valued as

$$k = \frac{2\pi}{\lambda} - i\alpha, \tag{19}$$

whose real and imaginary parts represent the wavenumber and the growth rate in the axial direction, respectively.

No-slip condition for liquid and gas at interface, in normal direction, reads as

$$U_l' = \frac{\partial \eta}{\partial z} + \frac{\partial \eta}{\partial t}, \tag{20a}$$

$$U_g' = \left(-2\sqrt{\frac{q}{l}} \sin\theta\right) \frac{\partial \eta}{r\partial \theta} + \frac{\partial \eta}{\partial t}, \tag{20b}$$

where η is the interface displacement, $\eta = r - 1$, and is expanded as

$$\eta = \sum_{m=-\infty}^{\infty} \hat{\eta}_m e^{im\theta} e^{i(kz-wt)}. \tag{21}$$

The dynamic balance of normal stress between gas and liquid reads as

$$\tau_l' - \tau_g' = -\frac{1}{We}\left(1 - k^2 - \frac{\sum_{m=-\infty}^{\infty} m^2 \hat{\eta}_m e^{im\theta}}{\sum_{m=-\infty}^{\infty} \hat{\eta}_m e^{im\theta}}\right)\eta, \tag{22}$$

where normal stresses are

$$\tau_l' = -p_l' + \left(\frac{2}{Re}\right)\frac{\partial U_l'}{\partial r}, \tag{23a}$$

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$$\tau_g' = -p_g' + \left(\frac{2h}{Re}\right)\frac{\partial U_g'}{\partial r}. \tag{23b}$$

The continuity of shear stress and tangential velocity across the jet boundary are dropped in irrotational approximations, consistent with viscous potential flow analysis. On the other hand, a KH instability that is due to velocity discontinuity cannot occur when the no-slip condition is applied.[22] The discrepancy between this shear stress and the zero shear stress required for exact solutions will generate a vorticity layer which is assumed to be very thin and, therefore, negligible in this analysis.

By substituting the normal modes into the governing equations, subtracting the base flow, and linearizing the resulting perturbation equations, the instantaneous equations describing the conservation of mass and momentum in the liquid and gas phases are obtained which can be written as

$$\nabla^2\phi_l' = 0, \quad \nabla^2\phi_g' = 0. \tag{24}$$

Following the method of separation of variables, the general solutions of (22) are then sought in the form of

$$\hat{\phi}_{l,m} = C_{1,m}I_m(kr) + C_{2,m}K_m(kr), \tag{25a}$$

$$\hat{\phi}_{g,m} = C_{3,m}I_m(kr) + C_{4,m}K_m(kr), \tag{25b}$$

where I_m and K_m are the mth order modified Bessel functions of the first and second kind, respectively. Since the liquid velocity is finite at $r = 0$ and the gas velocity should vanish as $r \to \infty$, coefficients of $C_{2,m}$ and $C_{3,m}$ must be equal to zero.

Through a tedious but straightforward calculation, by linearizing the kinematic and dynamic boundary condition Eqs. (20a), (20b), (23a), and (23b) using Eqs. (16), (17) and (21), multiplying them by $e^{-im\theta}/2\pi$, integrating the results from 0 to 2π, and substituting the inversion formulae in (16) and (17), a set of Fourier-transformed boundary conditions are obtained. By canceling $\hat{\eta}_m$ among these interface equations, they take the form of eigenvalue equations as

$$\begin{bmatrix} D_{11} & D_{12} \\ D_{21} & D_{22} \end{bmatrix}\begin{bmatrix} C_{1,m} \\ C_{4,m} \end{bmatrix} = 0, \tag{26}$$

where matrix components D_{11}, D_{12}, D_{21}, and D_{22} are

$$D_{11} = -i\omega I_m', \tag{27a}$$

$$D_{12} = -i(k-\omega)K_m', \tag{27b}$$

$$D_{21} = -i\omega\left[i(k-\omega)I_m + \frac{2}{Re}I_m''\right], \tag{27c}$$

$$D_{22} = l\omega^2 K_m + 2i\frac{h}{Re}\omega K_m'' - 12\frac{h}{Re}\sqrt{\frac{q}{l}}K_m' - \frac{1}{We}\left(1 - m^2 - k^2\right)K_m', \tag{27d}$$

in which primes denote the derivative with respect to r calculated at $r = 1$. Nontrivial solutions of the homogeneous Eq. (26), yield a dispersion relation. The roots of this equation in the desired parameter space will be studied in Sec. IV where for any arbitrary frequency, the corresponding growth rate and wavelength as the eigenvalue will be obtained.

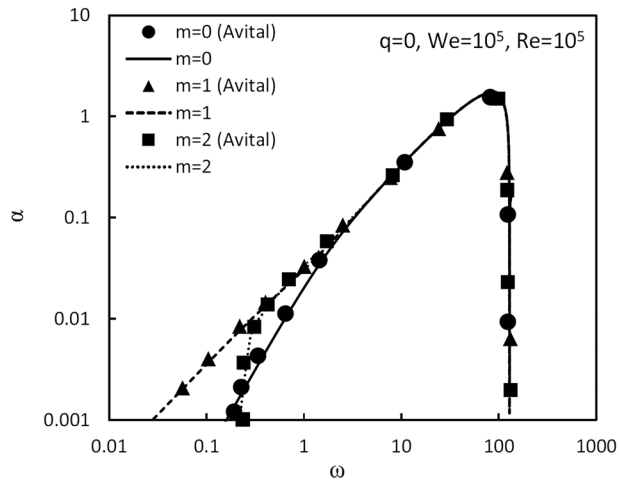

FIG. 5. Comparison of the instability range between present model and that of Avital.[13]

IV. RESULTS AND DISCUSSION

In the following, the characteristic equation obtained in Sec. III is used to investigate the effects of the aerodynamic force, surface tension, and viscosity on the growth rate and perturbation dynamics of liquid jets. We consider a range of operating conditions corresponding to capillary, first wind-induced, and second wind-induced regimes. Our primary interest is on air-water media with a density ratio of $l = 0.001\,18$ and a viscosity ratio of $h = 0.0208$. However, the effect of changing viscosity and density ratios will be considered as well using factors of these values, e.g., $h = 2\times$, $h = 3\times$, and $l = 2\times$, $l = 3\times$, indicating two-folds and three-folds of air-water ratios. The range of other parameters considered in this study is $10^3 < \mathrm{Re} < 10^4$ and $10^2 < \mathrm{We} < 10^4$. We consider a Reynolds number in the liquid phase such that it is high enough that the boundary layers are thin, and low enough such that the flow is non-turbulent. Results from the stability analysis are summarized in Figs. 5–10 for these selected cases.

To confirm the validity of the current approach, comparisons with classical linear stability results for the special case

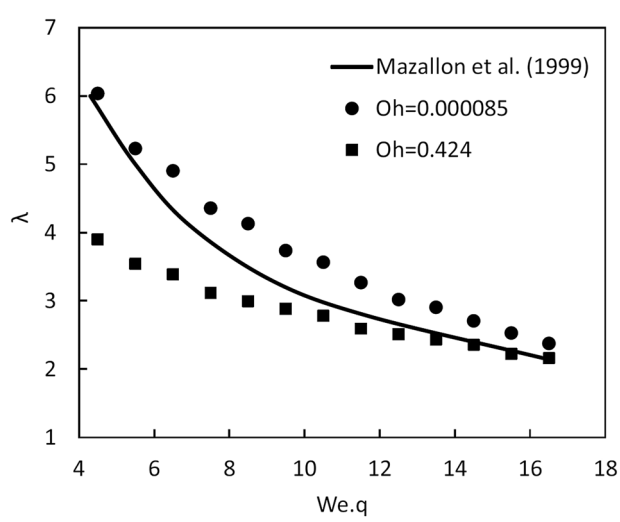

FIG. 6. Variation of the fastest growing wavelength with multiplication of Weber number and momentum ratio.

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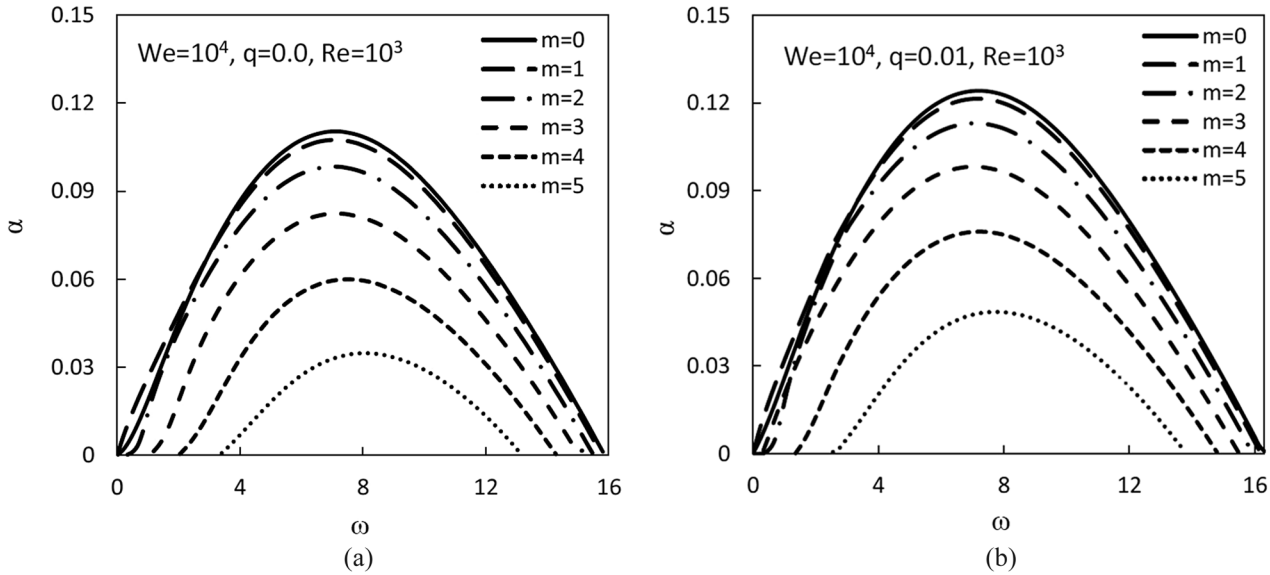

FIG. 7. Growth of perturbations at different azimuthal modes.

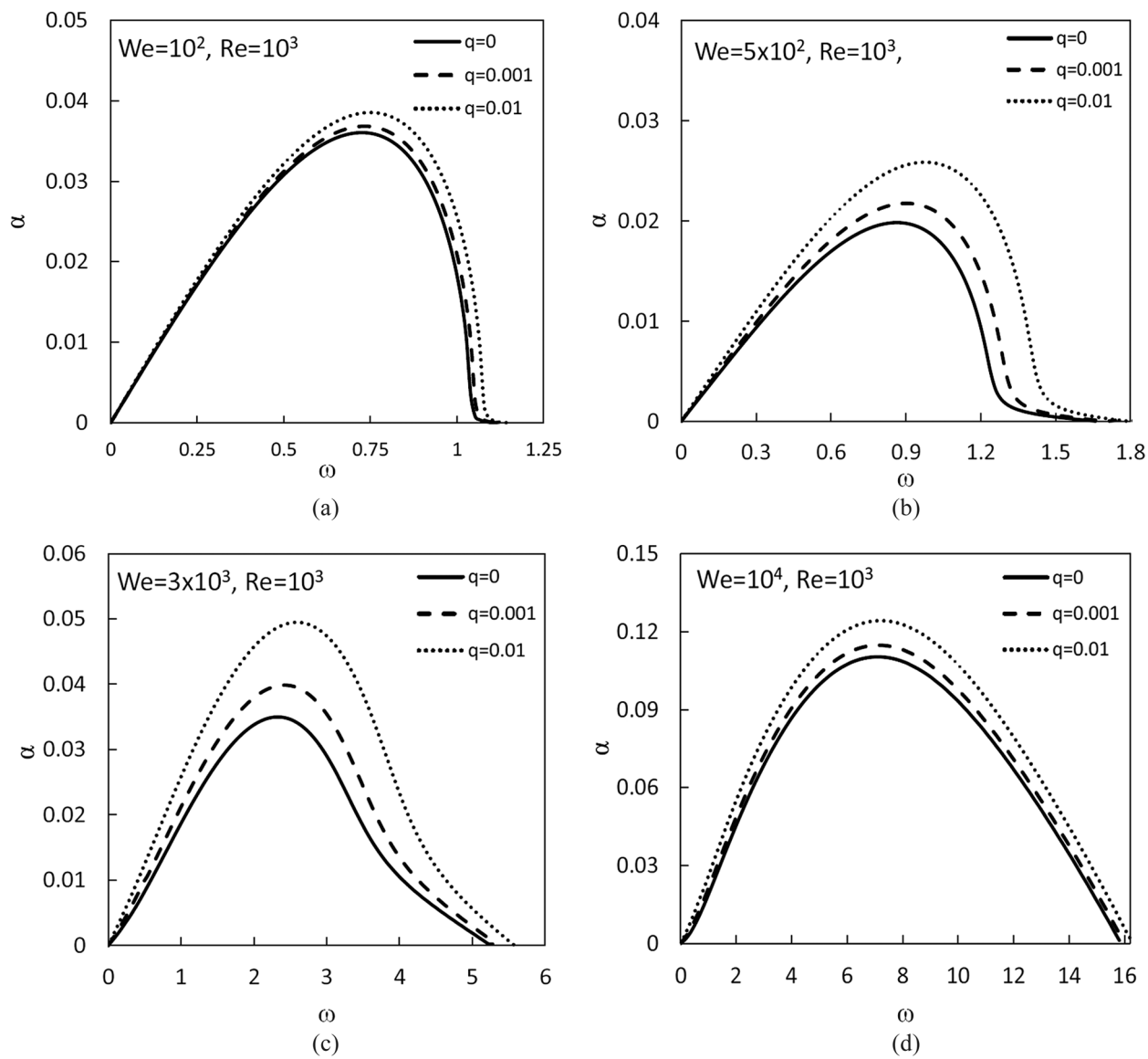

FIG. 8. Effect of crossflow momentum on growth rate at different breakup regimes.

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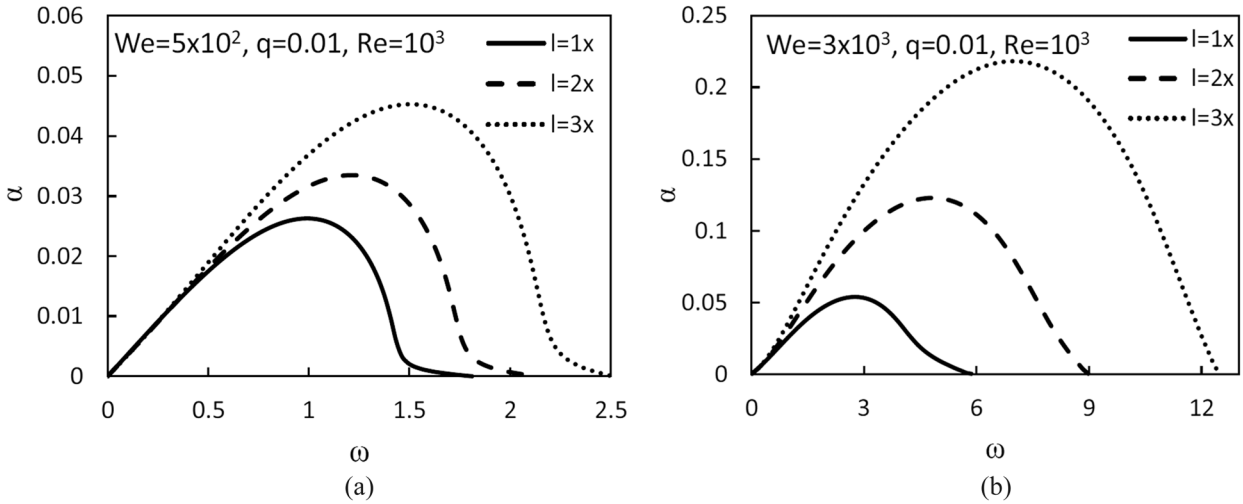

FIG. 9. Effect of crossflow-to-jet density ratio on instability.

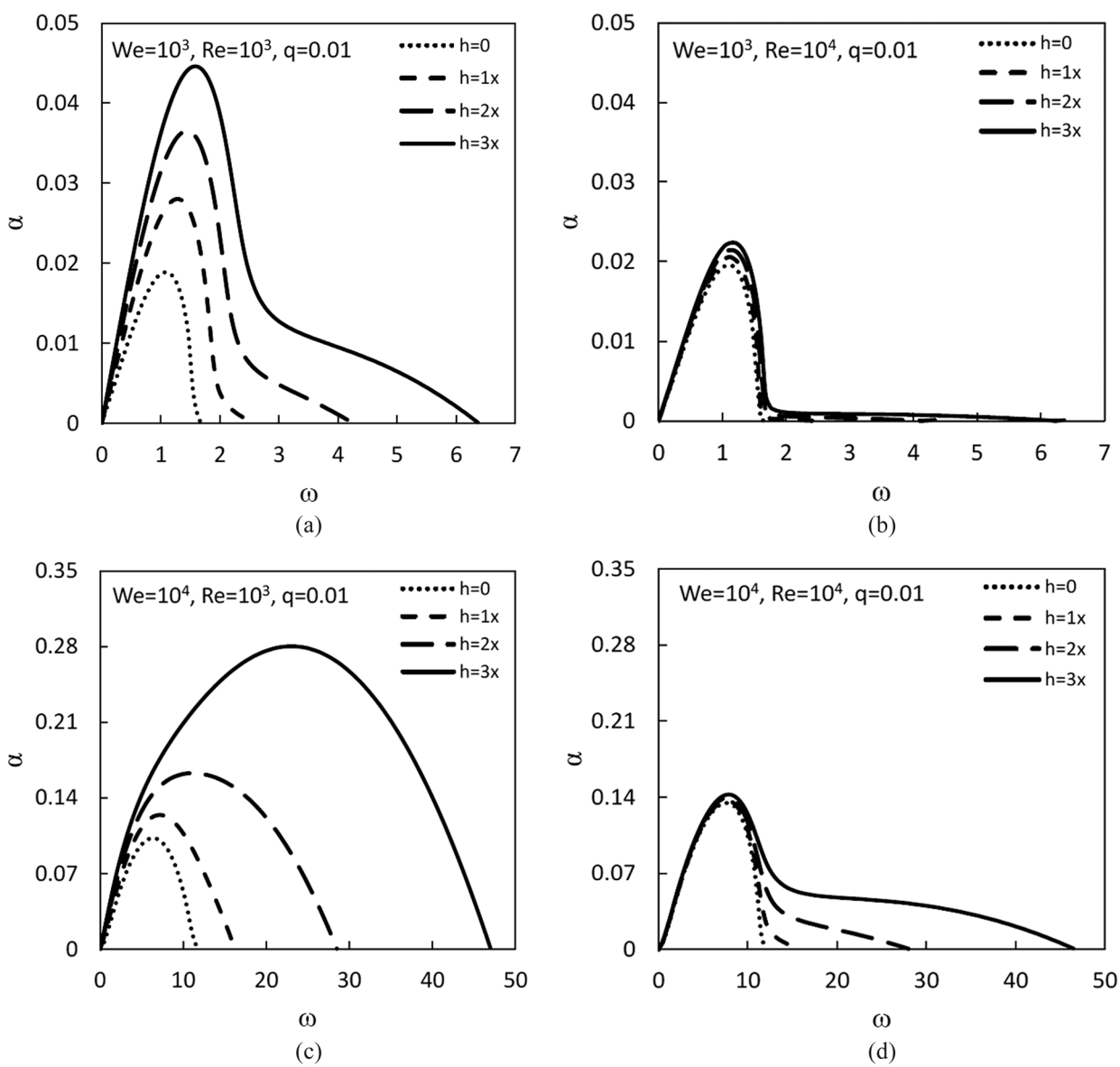

FIG. 10. Effect of crossflow-to-jet viscosity ratio on instability.

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of zero crossflow, i.e., free jets, have been made. For inviscid liquid and gas, the obtained dispersion Eq. (26) reduces to that of Yang.[12] For viscous liquid jet and inviscid crossflow, when $m = 0$, the dispersion equation reduces to that of Sterling and Sleicher.[11] When liquid jet is viscous, for the first three azimuthal modes, dispersion curves are comparable with classical results by Avital[13] as presented in Fig. 5. The excellent agreement serves as verification of the developed stability formulation.

The maximum growth rate and the corresponding wavenumber characterize the fastest growing waves on the liquid surface that are eventually responsible for the breakup. The associated wavelength, λ, along the liquid column should occur in a real physical problem and the ligament size is proportional to it. Following the current linear stability analysis, the maximum-growth wavelength is calculated from Eq. (26) and its dependence on the *We* and q at two different *Oh* is plotted in Fig. 6. This figure depicts the wavelength as a function of We·q, equivalent to the Weber number based on gas density in Eq. (1) associated with the crossflow as defined by Mazallon *et al.*[9] This figure also includes the correlation of Mazallon *et al.*[9] based on the measurements performed in the range of $0.000\,125 < q < 0.01$. In their experiment, for which one correlation has been offered as plotted in Fig. 6, different fluids with different viscosities were used. Although it was concluded that λ changes with *Oh* number, its dependency could not be resolved with their experimental uncertainties.[9] The present model, predicts the measured wavelength between minimum and maximum *Oh* numbers used in their experimental setup. The analysis predicts the same trend of decreasing wavelength with increasing *We* as the experimental results[29] and simulation results[6] show. As the product of *We* and q increases, the wavelength decreases for both values of *Oh*. As the *Oh* increases, also a decrease in wavelength is observed at all $We{\cdot}q$ values, suggesting that KH instability is more likely the controlling mechanism in the current range of operating conditions. We note that the capillary (Rayleigh-Plateau) instability always exist as a mechanism for the growth of waves.

The growth rate of instability waves as a function of the tangential wavenumber and momentum flux ratio can be calculated from Eq. (26). Figure 7 shows the growth rate at $We = 10^4$ and $Re = 10^3$, for different azimuthal modes when $q = 0$ and $q = 0.01$. It can be seen that for both cases, with and without crossflow, the axisymmetric mode has the highest growth rate and by increasing m growth rate decreases. Compared to a free jet, a liquid jet in crossflow has a higher growth rate while the same trend holds. For the rest of the results, emphasis has been placed on the axisymmetric mode $m = 0$ since the higher modes tend to have distinctly lower growth rates or be periodical. The amplitude of the columnar wave grows with increasing distance in the streamwise direction [demonstrated by Eq. (21)] and the jet breaks where the magnitude of the fluctuating part becomes equal to the mean jet radius, i.e., $\eta = 1$. These results are in agreement with experiments of Wu *et al.*[8] and Ng *et al.*[30] showing that the breakup length is reduced subjecting a liquid column to increasing crossflow momentum, resulting in the formation of smaller ligaments or droplets. It is noted that the actual degree of jet penetration depends strongly on the crossflow-to-jet momentum flux ratio.

The effect of crossflow momentum on the growth rate can be calculated also from Eq. (26). A comparison of disturbance growth rates for different Weber numbers when $Re = 10^3$ is presented in Fig. 8. This figure reveals that for all momentum ratios the instability region increases with increasing Weber number. It is noted that the Weber number of 10^2 belongs to the capillary regime, while Weber numbers of 10^3 and 10^4 belong to the first and second wind-induced regimes, respectively. Therefore, the intermediate *We* numbers will be associated with the transition between these regimes. The long-wave regime is dominant for low-speed (capillary regime) instabilities, while short waves are responsible for breakup in the case of high-speed jets. The instability growth rate decreases as the Weber number increases from 10^2 to 5×10^2 and then increases as *We* number goes to higher values. This behavior is due to capillary forces which have a destabilizing effect at the first wind-induced regime, while have a stabilizing effect at the second wind-induced regime. For all *We* numbers, it can be seen that the maximum growth rate becomes larger with increasing q and the peak of the dispersion curve shifts toward shorter waves. Furthermore, increasing the crossflow-to-jet momentum ratio leads to an extended range of unstable frequencies. It also shows that even in presence of RT instability, similar to KH instability alone, the growth of columnar waves are larger than those of azimuthal waves and the axisymmetric modes are more dominant.

Variation of the growth rate for different density ratios, as factors of that of air-water ratio, at $q = 0.01$ for $We = 5 \times 10^2$, 3×10^3 and $\mathrm{Re} = 10^3$, is shown in Fig. 9. From these results, it can be seen that by increasing crossflow-to-jet density ratio, the growth rate increases and wavelength of breakup reduces. This behavior is similar to that of a free jet which manifests the KH instability effects. As the density ratio is increased, the transition of dominance from KH to RT instability mechanisms on the liquid column surface is observed. Such a transition is consistent with the simulation results of Li and Soteriou.[7] The simulations by Herrmann[10] also showed a decrease in the surface wavelength with the increase in the density ratio. For the liquid jet in crossflow configuration, both instability mechanisms KH and RT can manifest themselves, and the competition between different mechanisms controls the development of surface waves and subsequent breakup of jet column. The jet destabilization and growth of column waves were predominantly attributed to a RT instability mechanism due to the presence of crossflow pressure gradient.

Another important parameter to study is the effects of viscosity of liquid jet and crossflow which are difficult to observe and measure experimentally. The importance is due to many applications in aerospace, pharmaceutical, and food industries.[1,6] Figure 10 shows variation of the growth rate for different gas-to-liquid viscosity ratios, as multiples of that of air-water at $We = \{10^3, 10^4\}$ and $Re = \{10^3, 10^4\}$. From these results, it can be seen that by increasing crossflow-to-jet viscosity ratio, the growth rate and instability range increase and the wavelength of the most dominant wave reduces. This means decreasing crossflow viscosity or increasing liquid jet viscosity damps the amplitude of instability waves also increases the

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characteristic length of such waves. These changes are more significant at lower *Re* numbers. The case of $h = 0$ represents the contribution of a pure KH instability where acceleration of an immiscible inviscid gas toward the liquid jet is absent. As the ratio of gas-to liquid viscosity is increased, a transition from the dominant KH instability to the dominant RT instability mechanisms on the liquid column surface occurs. This change in the instability is due to the viscosity of the crossflow which manifests itself in terms of RT disturbances.

V. CONCLUSIONS

A linear stability analysis was performed to examine the onset and growth of spatial instabilities of liquid jets in weak crossflows under consideration of surface tension, viscosity, and aerodynamic forcing. Providing a theoretical explanation for the flow behavior, the results show that aerodynamic forces associated with gas crossflow induce surface waves and accelerate the instability within the liquid column in the jet stream direction and cause column deformations which ultimately led to breakup. The results presented in this work reveal that by increasing crossflow-to-jet momentum ratio, the jet instability increases, the maximum growth rate shifts toward shorter waves, and the cut-off frequency does not change considerably. By increasing the viscosity or density ratio of crossflow-to-jet, the growth rate and cut-off frequency of instability increase. The axisymmetric instability has the highest growth rate among other azimuthal modes for the range of studied parameters. Although simplification assumption is made on jet deformation, the results are more comprehensive than a local stability analysis in which jet deformation is considered. The herein presented stability results provide access to instability behaviors characterized by the growth rate and wavenumber. The results could be of value in high-fidelity simulation codes[5–7] to adjust the complexity of the test cases. The numerical prediction of the growth rate and wavelength represents considerable challenges to numerical methods that can be systematically addressed through direct comparisons against theoretical results from the present analysis.

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